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M.Tech DEGREE EXAMINATION, MAY 2025

Second Semester

21CSC558J - DEEP LEARNING APPROACHES

(For the candidates admitted during the academic year 2024-2025)

Time: 3 Hours

Max. Marks: 100

(5 × 20 = 100 Marks)

Answer any 5 Questions

		Marks	BL	CO
1.	"You decide to use a multi-layer neural network with one hidden layer to solve the Ex-OR problem". Specify the activation function and number of neurons required in the hidden layer to solve the Ex-OR problem. Why is non-linearity important in this context?	20	3	1
2.	Training a neural network on a complex dataset shows high training and validation errors, indicating underfitting. Analyze potential reasons for underfitting related to model complexity, activation functions, and learning rate. What architectural modifications (e.g., number of layers, neurons, activation functions) would help reduce underfitting? How would increasing the learning rate affect the underfitting problem?	20	4	2
3.	Consider a 5×5 grayscale image and a 3×3 filter is given. Describe how the convolution operation is performed and calculate the size of the output feature map. Also, explain the role of stride and padding in determining the output size with the examples.	20	3	3
4.	"You are building a system to predict the next word in long English sentences". Justify how an LSTM helps solve the vanishing gradient problem and retains long-term dependencies. Describe the three gates of an LSTM with their function.	20	4	3
5.	Show the application of Variational auto-encoders in dimensionality reduction of data. Illustrate its differences from the denoising auto-encoder with the suitable example.	20	3	4
6.	Statement: "GANs enhance medical diagnostics by generating high-quality synthetic medical images". Elucidate the detailed architecture of GAN and its various types of networks used in applications.	20	4	4
7.	With a neat sketch, explain about the role of the Visual QA and Visual Dialog with the real time scenarios. And also describe the Pixel RNN for the learning process.	20	2	5
